1. (a) Chapter 8: Problem 16c

   (b) Chapter 8: Problem 21a,b

   (c) Chapter 8: Problem 27

   (d) Chapter 8: Problem 32

   (e) Chapter 8: Problem 47a,b,c.

Chapter 8, Problem 16(c), (d)

(c). Recall in Part (b) in Exercise set 1 we showed that $\widehat{\sigma}_{MLE} = \frac{1}{n}\sum_{i=1}^{n}|X_i|$. First, we write out the likelihood function for an simple observation X.

$$\begin{aligned}\text{lik}(\sigma) = \frac{1}{2\sigma}e^{-\frac{|x|}{\sigma}} &\implies \ell(\sigma) = -\log 2\sigma - \frac{|x|}{\sigma} \\ &\implies \ell'(\sigma) = -\frac{1}{\sigma} + \frac{|x|}{\sigma^2} \\ &\implies \ell''(\sigma) = \frac{1}{\sigma^2} - 2\frac{|x|}{\sigma^3}.\end{aligned}$$

Next, we note that

$$\mathbb{E}|X| = \int_{\mathbb{R}}|x|\,f(x\mid\sigma)\,dx = \int_{\mathbb{R}}\frac{|x|}{2\sigma}e^{-\frac{|x|}{\sigma}}dx = \int_0^{\infty}\frac{x}{\sigma}e^{-\frac{x}{\sigma}}dx = \mathbb{E}[Y] = \sigma,$$

where $Y \sim \text{Exp}\left(\frac{1}{\sigma}\right)$. Therefore, it follows that

$$I(\sigma) = -\mathbb{E}\left[\frac{\partial^2}{\partial\sigma^2}\log f(X|\sigma)\right] = -\mathbb{E}\left[\frac{1}{\sigma^2} - \frac{2|X|}{\sigma^3}\right] = \frac{2}{\sigma^3}\mathbb{E}[|X|] - \frac{1}{\sigma^2} = \frac{1}{\sigma^2}.$$

So the asymptotic variance of $\widehat{\sigma}_{MLE}$ is a.v.$[\widehat{\sigma}_{MLE}] = \frac{1}{nI(\sigma)} = \frac{\sigma^2}{n}$.

(d). Note that since

$$\begin{aligned}f(x_1,\ldots,x_n|\sigma) = \prod_{i=1}^{n}f(x_1|\sigma) &= \left(\frac{1}{2\sigma}\right)^n e^{-\frac{1}{\sigma}\sum_{i=1}^{n}|x_i|} \\ &= h(x_1,\ldots,x_n|\sigma)\,g(T(x_1,\ldots,x_n),\sigma),\end{aligned}$$

where $h(x_1,\ldots,x_n) = 1$ and $g(T(x_1,\ldots,x_n),\sigma) = \left(\frac{1}{2\sigma}\right)^n \mathrm{e}^{-\frac{1}{\sigma}T(x_1,\ldots,x_n)}$ and $T(x_1,\ldots,x_n) = \sum_{i=1}^{n}|x_i|$, then it follows from Factorization Theorem that a sufficient statistic for $X_1,\ldots,X_n$ can be $\sum_{i=1}^{n}|X_i|$.

Chapter 8, Problem 21

(a). First, we find the moment of X. Using integration by parts, one gets

$$\mathbb{E}[X] = \int_{\theta}^{\infty}xe^{-(x-\theta)}dx = \theta + 1 \Rightarrow \theta = \mathbb{E}[X] - 1.$$

Hence, $\widehat{\theta}_{MOM} = \overline{X} - 1$.

(b). We first write out the likelihood function as follows,

$$\text{lik}(\theta) = \prod_{i=1}^{n}\left(f(x|\theta)\,\mathbb{1}[x_i \geq \theta]\right)$$

$$
\begin{aligned}
&= e^{-\sum_{i=1}^{n}(x_i-\theta)} \prod_{i=1}^{n} \mathbb{1}\left[x_i \geq \theta\right] \\
&= e^{-\sum_{i=1}^{n}(x_i-\theta)} \mathbb{1}\left[x_i \geq \theta, \forall i = 1, \ldots, n\right] \\
&= e^{-\sum_{i=1}^{n}(x_i-\theta)} \mathbb{1}\left[\min x_i \geq \theta\right] \begin{cases} = 0 & \text{if } \theta > \min_i x_i \\ = e^{-\sum_{i=1}^{n}(x_i-\theta)} & \text{o.w} \end{cases}.
\end{aligned}
$$

Therefore, $\widehat{\theta}_{MLE} \leq \min_i X_i$, since otherwise the likelihood function will be zero. Now we show that $\widehat{\theta}_{MLE} = \min X_i$. Suppose on the other hand, $\widehat{\theta}_{MLE} = \min X_i - \varepsilon$, for some $\varepsilon > 0$. Then

$$
\frac{\operatorname{lik}\left(\widehat{\theta}_{MLE}\right)}{\operatorname{lik}\left(\min X_i\right)} = e^{-\left(\sum_{i=1}^{n} x_i - \min X_i + \varepsilon\right) + \left(\sum_{i=1}^{n} x_i - \min X_i\right)} = e^{-\varepsilon} < 1.
$$

So $\operatorname{lik}(\widehat{\theta}_{MLE}) < \operatorname{lik}\left(\min X_i\right)$, and therefore $\widehat{\theta}_{MLE} = \min X_i$.

(c). Notice that

$$
\begin{aligned}
f\left(x_1, \ldots, x_n|\theta\right) &= \prod_{i=1}^{n} f\left(x_i|\theta\right) \mathbb{1}\left[x_i \geq \theta\right] \\
&= e^{-\sum_{i=1}^{n}(x_i-\theta)} \mathbb{1}\left[\min x_i \geq \theta\right] \\
&= e^{-\sum_{i=1}^{n} x_i} e^{n\theta} \mathbb{1}\left[\min x_i, \geq \theta\right] \\
&= g\left(T\left(x_1, \ldots, x_n\right), \theta\right) h\left(x_1, \ldots, x_n\right),
\end{aligned}
$$

where $g\left(T\left(x_1, \ldots, x_n\right), \theta\right) = e^{n\theta} \mathbb{1}\left[T\left(x_1, \ldots, x_n\right) \geq \theta\right]$ and $T\left(x_1, \ldots, x_n\right) = \min_i x_i$ and $h\left(x_1, \ldots, x_n\right) = e^{-\sum_{i=1}^{n} x_i}$. Therefore, by Factorization Theorem $\min X_i$ is a sufficient statistic for $X_1, \ldots, X_n$

Chapter 8, Problem 27

(a). Let $X_1, \ldots, X_5$ denote the failing time of the 5 components. By the conditions given in the problem, the data given to us is $V = \min_{i=1}^{5} X_i$. By Example A in Section 3.7, we see that $V \sim \operatorname{Exp}\left(\frac{5}{\tau}\right)$. Therefore, the likelihood function is

$$
\operatorname{lik}\left(\tau\right) = f\left(v_1|\tau\right) = \frac{5}{\tau} e^{-\frac{5}{\tau} v_1},
$$

where $v_1 = 100$.

(b). First, we compute the log-likelihood function,

$$
\ell\left(\tau\right) = \log 5 - \log \tau - \frac{5v_1}{\tau} \implies \ell'\left(\tau\right) = -\frac{1}{\tau} + \frac{5v_1}{\tau^2}.
$$

Setting $\ell'\left(\tau\right) = 0$ yields $\widehat{\tau}_{MLE} = 5V$, which evaluated at $V = 100$ yields 500.

(c). We find out the distribution of $\widehat{\tau}_{MLE}$ by looking at its cdf. Note

$$F_{\widehat{\tau}_{MLE}}(t) = \mathbb{P}(\widehat{\tau}_{MLE} \leq t) = \mathbb{P}(5V \leq t) = \mathbb{P}\left(V \leq \frac{t}{5}\right)$$
$$= 1 - e^{-\frac{5}{\tau}\cdot\frac{t}{5}} = 1 - e^{\frac{-t}{\tau}} = F_Z(t),$$

where $Z \sim \text{Exp}\left(\frac{1}{\tau}\right)$. Hence, $\widehat{\tau}_{MLE} \sim \text{Exp}\left(\frac{1}{\tau}\right)$.

(d). Since we have in the previous part that $\widehat{\tau}_{MLE} \sim \text{Exp}\left(\frac{1}{\tau}\right)$, it follows that

$$\text{s.e}\left[\widehat{\tau}_{MLE}\right] = \sqrt{\text{Var}\left[\widehat{\tau}_{MLE}\right]}\Big|_{\tau=\widehat{\tau}_{MLE}} = \tau|_{\tau=\widehat{\tau}_{MLE}} = 500.$$

Chapter 8, Problem 32

(a) We have seen that in the case of a normal distribution with μ and σ^2 unknown, $\widehat{\mu}_{MLE} = \overline{X}$ and $\widehat{\sigma^2}_{MLE} = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$. We use these two as our estimates.

```
# form the data
data = c(5.3299,4.2537,3.1502,3.7032,1.6070,6.3923,3.1181,
         6.5941,3.5281,4.7433,0.1077,1.5977,5.4920,1.7220,
         4.1547,2.2799)
alpha = c(0.9,0.95,0.99)

# get lenth
n = length(data)

# compute MLE
mu_MLE = mean(data)
sigma_MLE = (n-1/n)*sd(data)

mu_MLE

## [1] 3.610869

sigma_MLE

## [1] 29.46538
```

(b). We apply the formula given in the textbook in page 280 for the confidence interval:$a\mathbb{P}$

$$\left(\overline{X} - \frac{S}{\sqrt{n}}t_{n-1}\left(\frac{\alpha}{2}\right), \overline{X} + \frac{S}{\sqrt{n}}t_{n-1}\left(\frac{\alpha}{2}\right)\right) \text{ for } \mu,$$
$$\left(\frac{n\widehat{\sigma}^2}{\chi^2_{n-1}(\alpha/2)}, \frac{n\widehat{\sigma}^2}{\chi^2_{n-1}(1-\alpha/2)}\right) \text{ for } \sigma^2$$

where $t_{n-1}(\alpha/2)$ denotes the point beyond which the t-distribution with $n-1$ degrees of freedom has probability $\alpha/2$. The code is as follows: a

```
# form the data
data = c(5.3299,4.2537,3.1502,3.7032,1.6070,6.3923,3.1181,
         6.5941,3.5281,4.7433,0.1077,1.5977,5.4920,1.7220,
         4.1547,2.2799)
alpha = c(0.9,0.95,0.99)

# get lenth
n = length(data)

# compute MLE
mu_MLE = mean(data)
sigma_MLE = (n-1/n)*sd(data)

# compute the CI's for mu
CI_mu = list()
for (i in alpha){
  lower = mu_MLE + sigma_MLE/sqrt(n) * qt(0.5-i/2,n-1)
  upper = mu_MLE + sigma_MLE/sqrt(n) * qt(0.5+i/2,n-1)
  CI_mu = c(CI_mu,list(c(lower,upper)))
  names(CI_mu)[which(alpha == i)] = paste(toString(i*100),
                                          "% ","confidence CI",sep= "")
}
CI_mu

## $`90% confidence CI`
## [1] -9.302706 16.524443
##
## $`95% confidence CI`
## [1] -12.09013  19.31186
##
## $`99% confidence CI`
## [1] -18.09564  25.31737

# compute the CI's for sigma^2
sigma2 = sigma_MLE^2
CI_sigma2 = list()
for (i in alpha){
  upper = sigma2 * n / qchisq((1-i)/2,n-1)
  lower = sigma2 * n / qchisq(1-(1-i)/2,n-1)
  CI_sigma2 = c(CI_sigma2,list(c(lower,upper)))
  names(CI_sigma2)[which(alpha == i)] = paste(toString(i*100),"% ",
                                              "confidence CI",sep= "")
```

```
}
CI_sigma2

## $`90% confidence CI`
## [1]  555.7472 1913.1590
##
## $`95% confidence CI`
## [1]  505.3529 2218.3063
##
## $`99% confidence CI`
## [1]  423.4994 3019.2556
```

(c). In page 280, the textbook showed that

$$\mathbb{P}\left(\frac{n\widehat{\sigma^2}_{MLE}}{\chi^2_{n-1}(\alpha/2)} \leq \sigma^2 \leq \frac{n\widehat{\sigma^2}_{MLE}}{\chi^2_{n-1}(1-\alpha/2)}\right) = \mathbb{P}\left(\sqrt{\frac{n\widehat{\sigma^2}_{MLE}}{\chi^2_{n-1}(\alpha/2)}} \leq \sigma \leq \sqrt{\frac{n\widehat{\sigma^2}_{MLE}}{\chi^2_{n-1}(1-\alpha/2)}}\right) = 1-\alpha.$$

Hence, we use this to get our confidence interval for σ.

```
CI_sigma = lapply(CI_sigma2,sqrt)
CI_sigma

## $`90% confidence CI`
## [1] 23.57429 43.73967
##
## $`95% confidence CI`
## [1] 22.48006 47.09890
##
## $`99% confidence CI`
## [1] 20.57910 54.94775
```

(d). We use the 95% confidence interval as an example. We have the length of this interval as $(2S/\sqrt{n})t_{n-1}(\alpha/2)$. We use the upper point, denoted as $\sigma^{99\%}_{upper}$ of the 99% confidence interval to replace S to make a most conservative estimate. So we would like to find n s.t. $\left(2\sigma^{99\%}_{upper}/\sqrt{n}\right)t_{n-1}(\alpha/2) \leq (S/\sqrt{n})\,t_{n-1}(\alpha/2)$ for $\alpha = 0.95$. An exhaustive search yields $n = 86$.

```
# find n that halves the interval
sigma_max_estimate = unlist(CI_sigma[3],use.names = F)[2]
sigma_max_estimate

## [1] 54.94775

length = unlist(CI_mu[2],use.names = F)[2] - unlist(CI_mu[2],use.names = F)[1]
```

```
while(2 * sigma_max_estimate/sqrt(n) * qt(0.5+i/2,n-1) >= length){
  n=n+1
}
n

## [1] 86
```

Chapter 8, Problem 47

(a). First, we find the moment of X_1.

$$\begin{aligned}\mathbb{E}[X_1] &= \int_{x_0}^{+\infty} x f(x|x_0,\theta)\,dx = \int_{x_0}^{+\infty} \theta x_0^{\theta} x^{-\theta} dx \\ &= \theta x_0^{\theta} \lim_{t\to\infty} \left[\frac{x^{-\theta+1}}{1-\theta}\right]_{x_0}^{t} = \theta x_0^{\theta}\left[0 - \frac{x_0^{1-\theta}}{1-\theta}\right] = \frac{\theta}{\theta-1}x_0.\end{aligned}$$

Rearranging yields

$$\begin{aligned}\frac{x_0}{\mathbb{E}[X]} = 1 - \frac{1}{\theta} &\implies \frac{1}{\theta} = 1 - \frac{x_0}{\mathbb{E}[X]} \\ &\implies \theta = \frac{1}{1-\frac{x_0}{\mathbb{E}[X]}} = \frac{\mathbb{E}[X]}{\mathbb{E}[X]-x_0}.\end{aligned}$$

Therefore, $\widehat{\theta}_{MOM} = \frac{\overline{X}}{\overline{X}-x_0}$.

(b). First, we write out the likelihood equations for θ as usual:

$$\begin{aligned}\text{lik}(\theta) &= \prod_{i=1}^{n} f(x_i|\theta,x_0) = \theta^n x_0^{n\theta} \prod_{i=1}^{n} x_i^{-\theta-1} \\ \implies \ell(\theta) &= n\log\theta + n\theta\log x_0 - (\theta+1)\sum_{i=1}^{n}\log x_i \\ \implies \ell'(\theta) &= \frac{n}{\theta} + n\log x_0 - \sum_{i=1}^{n}\log x_i.\end{aligned}$$

Setting ℓ' to zero yields

$$\begin{aligned}\ell'(\theta) = 0 &\iff \frac{n}{\theta} + n\log x_0 - \sum_{i=1}^{n}\log x_i = 0 \\ &\iff \frac{1}{\theta} = \frac{1}{n}\sum \log x_i - \log x_0 \\ &\iff \theta = \frac{n}{\sum_{i=1}^{n}\log x_i - n\log x_0}.\end{aligned}$$

Hence, $\widehat{\theta}_{MLE} = \frac{n}{\sum_{i=1}^{n}\log X_i - n\log x_0}$.

(c). We consider the Fisher Information for one single observation, X_1. First, note that $\ell''(\theta) = -\frac{1}{\theta^2}$. Then

$$I(\theta) = -\mathbb{E}[\ell''(\theta)] = -\mathbb{E}\left[-\frac{1}{\theta^2}\right] = \frac{1}{\theta^2}.$$

Therefore,

$$\text{a.v.}[\widehat{\theta}_{MLE}] = \frac{1}{nI(\theta)} = \frac{1}{n/\theta^2} = \frac{\theta^2}{n}.$$

(d). Note that by independence

$$\begin{aligned} f(x_1, \ldots, x_n|\theta, x_0) &= \prod_{i=1}^{n} f(x_i|\theta, x_0) = \theta^n x_0^{n\theta} \prod_{i=1}^{n} x_i^{-(\theta+1)} = \theta^n x_0^{n\theta} \left(\prod_{i=1}^{n} x_i\right)^{-(\theta+1)} \\ &= h(x_1, \ldots, x_n)\, g(T(x_1, \ldots, x_n), \theta), \end{aligned}$$

where $g(T(x_1, \ldots, x_n), \theta) = \theta^n x_0^{n\theta} [T(x_1, \ldots, x_n)]^{-(1-\theta)}$ and $T(x_1, \ldots, x_n) = \prod_{i=1}^{n} x_i$ and $h(x_1, \ldots, x_n) = 1$. Hence, it follows by Factorization Theorem that a sufficient statistic for $X_1, \ldots, X_n$ can be $\prod_{i=1}^{n} X_i$.